



Tecnológico de Monterrey

INSTITUTO TECNOLÓGICO Y DE ESTUDIOS
SUPERIORES DE MONTERREY

CAPSTONE PROJECT (TEAM #49)

Project Proposal

Author Julia Gabriela Pinedo A01795315

Professor Dr. Grettel Barceló Alonso

Project Advisor Dr. Guillermo Mota Medina

21.09.2025

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1 Project Background

The **box-pushing problem** has become a crucial benchmark task in cooperative robotics, as it models scenarios where one or more robots must push a heavy box toward a goal region while avoiding obstacles and walls. [1] This task demands superior coordination between the robots, which has led to its adoption in **Multi-Agent Reinforcement Learning (MARL)** as a way to improve efficiency and reduce the risk of counterproductive actions while navigating towards the designated goal. MARL is the paradigm for fully cooperative tasks where agents must share intent, avoid interference, and learn complementary roles. Recent benchmarking works explicitly include box-pushing among canonical cooperative MARL tasks, underscoring its value for evaluating coordination and robustness. [2][3]

Although the present study is **academic in scope**, the importance of solving this problem extends across multiple domains, including logistics, warehouse automation, and search-and-rescue operations, where collective transport of heavy or bulky objects is required. Therefore, research has progressed from centralized controllers and leader-follower strategies to decentralized and learning-based methods, aiming to improve robustness, scalability, and autonomy. [1][4]

Latest findings suggest that naive MARL approaches could underperform compared to *single-agent solutions* due to issues such as coordination failures and credit assignment problems, motivating the need for an approach that can significantly enhance cooperative learning without heavily depending on environments too complex to be fully modeled or strategies with heavy computational costs. [5] This context motivates the development of a project that not only establishes a single-agent reinforcement learning baseline but also advances toward a robust multi-agent solution capable of addressing these limitations.

1.1 Company / Institution

Tsinghua University, founded in 1911 and located in northwest Beijing, is one of China's most prestigious and influential universities. It has grown into a comprehensive research university with 22 schools and 59 departments in science, engineering, humanities, and more. [6]



清华大学 车辆与运载学院

School of Vehicle and Mobility, Tsinghua University

Figure 1.1 Logo of the institution

Tsinghua's **School of Vehicle and Mobility** is a department dedicated to cultivating high-level talent in automotive engineering, intelligent vehicles, and mobility systems [7]. Its research agenda emphasizes intelligent transportation systems, autonomous driving technologies, and collaborative robotics, positioning it as a *highly relevant* institution for a project focused on Multi-Agent Reinforcement Learning.

This project aligns directly with the *school's purpose*, as box-pushing is a canonical cooperative robotics benchmark within autonomous mobility. Addressing this task contributes directly to the school's broader objectives of advancing autonomous mobility research and developing innovative solutions for future intelligent vehicle and robotic systems.

1.2 Business Processes Impacted

According to the SCIAN 2023 classification [8], the project fits in the **(54) Professional, Scientific, and Technical Services** category. This classification denotes that this project is going to primarily impact the area of **scientific research** and **experimental development** in robotics and artificial intelligence, specifically in the area of MARL for collaborative robotics. By simulating and analyzing box-pushing scenarios, the project directly contributes to:

- Advancing methodologies in collaborative robotics, where multiple autonomous agents must coordinate in dynamic environments.
- Improving collaborative robotics frameworks relevant to logistics, warehouse automation, and smart factory applications, where multi-robot teams are increasingly used for material handling.
- Enabling robust design of autonomous decision-making systems, providing insights into strategies that make future robotic solutions more scalable, efficient, and resilient.

In this sense, while the immediate scope is academic, the *long-term impact* of the project extends to industries that depend on collective transport and autonomous mobility. By addressing fundamental coordination challenges, the project reinforces Tsinghua University's role in bridging research with real-world intelligent systems development.

2 Business Understanding

2.1 Problem Definition

Multi-robot formation control has been extensively studied because of its importance in scenarios where multiple robots must collaborate to move a payload that is too heavy or bulky for one robot. In such *self-organizing systems*, the agents need to operate in coordination to avoid conflicts such as inefficient paths, counterproductive forces that result in little or no movement, or impeded learning progress. Since this problem involves one or more robots working together, it is connected to the *multi-robot box pushing problem*, a classic cooperative multi-agent task in robotics, where one or more robots must push a box toward a goal region while avoiding obstacles. [9][10]

The key challenges of this problem include:

- Enabling robots to find efficient paths and coordinate their pushes.
- Planning and executing rotational maneuvers to avoid obstacles in the environment.
- Supporting adaptive decision-making as the “*state*” (positions of robots and box) continuously changes.

Although multi-agent solutions are conceptually more scalable and robust, in practice, single-agent reinforcement learning approaches have often demonstrated superior performance in terms of efficiency and reliability, and simple MARL implementations typically have limitations. This gap **underscores the need to enhance cooperative learning in multi-robot formation and box-pushing tasks**. Establishing a single-agent reinforcement learning baseline is essential, both to highlight the limitations of the current MARL methods and to provide a clearer benchmark for developing more robust multi-agent solutions.

2.2 Context

Addressing the box-pushing problem through a Multi-Agent Reinforcement Learning approach is essential, as it represents a **fundamental challenge in collaborative robotics** and provides insights that extend beyond the multi-robot benchmark itself. From the academic perspective, it serves as a rigorous experiment for evaluating coordination strategies in MARL. Demonstrating a successful performance in this domain can provide a *scalable design* where multiple robots could outperform a single agent, advancing the state of the art in artificial intelligence and reinforcement learning research.

From a real-world perspective, the ability of multiple autonomous robots to collaborate efficiently has its roots in areas such as logistics, warehouse automation, and manufacturing, where teams of robots could be deployed to handle material transport tasks. In these contexts, inefficient coordination leads to higher energy costs, longer processing times, and reduced system reliability. Moreover, solving this problem can also be relevant to disaster response and search-and-rescue operations, where autonomous multi-robot teams are required to clear debris or heavy obstacles under uncertain and unpredictable conditions.

Although the immediate focus of the project is academic experimentation, the problem remains significant because it originates from real-world needs and contributes to the *long-term development* of more resilient, efficient, and autonomous robotics solutions capable of operating in complex scenarios or environments.

2.3 Objectives

The project has the following objectives:

- Establish a single-agent reinforcement learning baseline by implementing **Advantage Actor-Critic (A2C)** and **Proximal Policy Optimization (PPO)**.

These algorithms will be compared in terms of performance and efficiency, providing a clear reference point for evaluating more complex approaches.

- Extend the same **baseline algorithms** (*A2C and PPO*) to a multi-agent setting.

This benchmark will serve as a foundation for identifying the strengths and weaknesses of standard MARL approaches.

- Design and implement a **robust multi-agent RL framework** that enables effective cooperation among multiple differential-drive robots in simulated environments.
- Integrate a **custom differential-drive robot** model equipped with odometry, IMU, and contact detection sensors to support a realistic RL environment in Gazebo + ROS2. [11][12]
- Evaluate performance across **key metrics**.

This will include: success rate, efficiency (*steps to reach the goal*), robustness under agent failures, and adaptability to obstacles.

2.4 Key Questions

The project aims to address the following questions:

- In what ways can extending A2C and PPO to the multi-agent setting establish a fair benchmark for analyzing coordination, efficiency, and robustness in the box-pushing problem?
- What reward structures, state representations, and action spaces most effectively enable multiple robots to cooperate and avoid counterproductive behaviors, such as inefficient paths or sliding due to misaligned pushes?
- How can a MARL framework be designed to remain robust under dynamic test conditions, including larger environments, the presence of multiple obstacles, and potential agent failures?
- How can a single-agent RL baseline provide benchmark data that highlights the gaps and improvements achieved by a Multi-Agent Reinforcement Learning approach?

2.5 Stakeholders

The project involves a small but focused team within the academic context of Tsinghua University's School of Vehicle and Mobility:

Academic Stakeholders

- *Dr. Fuxi Wen, Associate Professor.* He serves as an academic advisor, providing supervision and methodological guidance throughout the project.

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- *Julia Gabriela Pinedo, Master's Student and Mechatronics Engineer.* She is responsible for the implementation of the simulation, the design of the reinforcement learning framework, and the analysis of experimental results.

Technical Components

- ROS2, Gazebo, Visual Studio, GitHub.

This is an academic collaboration at the stage of prototype and experimentation, intending to advance academic understanding of MARL rather than an immediate industrial application.

3 Data Understanding

3.1 Data Description

This project does not rely on a traditional static dataset. Instead, the data is generated through interaction with the simulation environment in Gazebo + ROS2. The environment consists of:

- A **differential-drive robot model** with defined physical dimensions, sensors (odometry, IMU, contact detection), and control inputs.

Table 3.1 Dimensions of the robot.

Part	Dimensions
Base (main body)	L: 0.35 m W: 0.30 m H: 0.12 m Mass: ~ 4.5 kg
Wheels (2x)	Radius: 0.08 m Width: 0.03 m Mass: 0.3 kg
Wheel separation	0.30 m
Caster	Radius: 0.03 m Mass: 0.2 kg
Pusher plate	Thick: 0.02 m Wide: 0.25–0.30 m High: 0.10–0.12 m Mass: 0.5 kg

- A **box** with fixed dimensions and mass, serving as the shared object to be pushed.

Table 3.2 Dimensions of the box.

Part	Dimensions
Box (object)	0.50 m $\times$ 0.50 m $\times$ 0.50 m Mass: ~ 5 kg

- A **simulation arena**, including floor surface properties (friction coefficient) and placement of static obstacles.

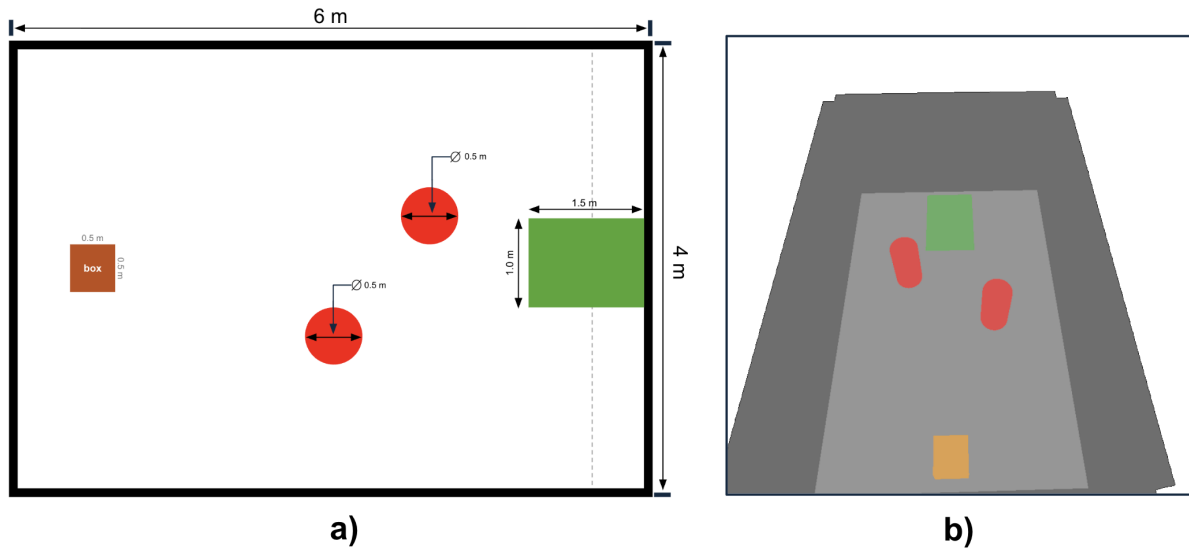


Figure 3.1 Simulation environment. a) Diagram of the arena, b) Arena modeled and simulated in Gazebo.

3.2 Machine Learning Technique

The project applies **Deep Reinforcement Learning (DRL)** methods. Unlike classical supervised or unsupervised learning, which rely on static datasets, DRL focuses on *learning through interaction*. An agent observes the current state of the environment, selects an action, receives a reward, and transitions to a new state. Over time, the agent improves its decision-making policy to maximize cumulative reward. [13]

This trial-and-error paradigm is well-suited for robotics and control problems, where data is not available beforehand but instead generated through simulated or real interactions. DRL is more suitable than supervised or unsupervised techniques because the learning process relies on sequential actions and feedback loops, rather than static input-output mappings. [14]

The DRL methods applicable to this project are the following:

Advantage Actor-Critic (A2C)

This combines **policy-based** and **value-based learning**. The actor learns a parameterized policy that maps states to actions, while the critic learns a value function to estimate how good a state is under the current policy [15]. The key update in A2C is driven by the advantage function:

$$A(s_t, a_t) = R_t + \gamma V(s_{t+1}) - V(s_t)$$

where:

- R_t is the reward,
- γ is the discount factor,
- $V(s_t)$ is the estimated value of the current state.

The policy gradient update is then:

$$\nabla_{\theta} J(\theta) = \mathbb{E}[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) A(s_t, a_t)]$$

This means that the actor increases the probability of actions that yield positive advantages and decreases it for actions with negative advantages.

Proximal Policy Optimization (PPO)

This improves upon standard policy gradient methods by inducing a **clipping mechanism** that prevents overly large policy updates, which can destabilize training. [16]

The PPO objective is:

$$L_{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right]$$

where:

- $r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$ is the probability ratio between new and old policies,
- A_t is the advantage estimate,
- ϵ is a small constant (*e.g.*, 0.1, 0.2) controlling the clipping range.

The clipping ensures that if the new policy diverges too much from the old one, the update is restricted. This stabilizes learning and often yields state-of-the-art performance in continuous control tasks.

3.3 Variable Identification

In this context, the variables correspond to **states**, **actions**, and **rewards**.

Inputs (state variables):

- Robot odometry (x, y [m], yaw [rad])
- IMU yaw rate [rad/s]
- Contact detection flag [0/1]
- Box position [m], orientation [rad]
- Relative position of the goal [m]
- Obstacle positions [m]

Outputs (action variables):

- Linear velocity v [m/s]
- Angular velocity ω [rad/s]
- Direct wheel velocities [rad/s]

Reward signal (feedback):

- Positive reward for reducing the distance between the box and the goal
- Penalties for collisions, force cancellation (*sliding/inefficient pushes*), or excessive rotations
- Strong reward when the box reaches the goal region

4 References

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5 Attachments

Signed document of the collaboration between Tecnológico de Monterrey and Tsinghua University for this project.



COLLABORATION COMMITMENTS FOR THE CAPSTONE PROJECT

School of Engineering and Sciences

Master in Applied Artificial Intelligence

MNA

GENERAL

The Capstone Project corresponds to the final project carried out in the Master's program in Applied Artificial Intelligence. This will be developed and evaluated in the course with code TC5035, of the same name.

The main objective of the project is to solve a real, current problem, whether in the industrial sector or government institutions, NGOs, or some other type of institution. The above, prior authorization of the committee.

For the purposes of this document, the Master's program in Applied Artificial Intelligence (hereinafter referred to as MNA) and Tsinghua University (hereinafter referred to as 'the client') acknowledge the following commitments:

COMMITMENTS

To carry out the Capstone Project, the MNA commits to the following:

1. Assign a project advisor with experience and expertise to the problem presented by the client.
2. Provide the necessary facilities and existing equipment that contribute to solving the identified problem.
3. Offer constant and systematic feedback during the proposal registration and project development in the TC5035 course.
4. Form the evaluation committee once the objectives presented in the Capstone Project proposal have been achieved.

The client commits to:

1. Assign a sponsor with recognized strengths in the domain area to be addressed by the project.
2. Provide feedback to the project member(s) regarding the partial results obtained.
3. Ensure the information resources required for the project's completion.
4. Recognize the participation of the student(s) as well as the Tecnológico de Monterrey in any publications or reports generated from the project.

Both institutions commit to:

1. Maintain confidentiality in relation to the results obtained that are sensitive/strategic for any of the institutions.

2. Two-way interaction in work meetings to optimize the implementation process of the Capstone Project.

The collaboration will take place (without limitation) for the implementation of the “**Multi-Agent Reinforcement Learning for Collaborative Box-Pushing**” project, starting from September 8th, and tentatively concluding on November 21th, 2025.

Capstone Project Student(s):



Julia Gabriela Pinedo, A01795315

Name, registration number and signature.

Director of the MNA:



Dr. Luis Eduardo Falcón, National Director of the Master in Applied Artificial Intelligence

Name, position and signature.

Sponsor assigned by client:



Dr. Fuxi Wen. Associate Professor in School of Vehicle and Mobility, Tsinghua University

Name, position and signature.